

Student in Financial Mathematics at Paris Dauphine University. Currently seeking an end-of-studies internship.

Education and Degrees

- 2024– **Master's degree ISF (Statistical and Financial Engineering)**, Paris-Dauphine University, Paris, FRANCE
- 2018–2019 **Master's degree MVA (Mathematics, Vision, Machine Learning) , with highest honors**, *Ecole Normale Supérieure de Paris-Saclay*, Cachan, FRANCE
- 2016–2020 **Ecole Normale Supérieure de Paris-Saclay (Admission by Competitive Exam)**, *Fundamental and Applied Mathematics, Machine Learning*, Cachan, FRANCE
- 2015–2016 **Bachelor's Degree in Mathematics, Third Year, with Honors**, *Pierre et Marie Curie University*, Paris, FRANCE
- 2012–2015 **Preparatory classes (MPSI, MP, MP*) for competitive entrance exams to the Grandes Écoles**, *Lycée Saint-Louis*, Paris, FRANCE
- 2012 **Scientific Baccalaureate with highest honors**, *Jean Moulin High School*, Saint-Amand Montrond, FRANCE

Research Internships

- 05/2021–08/2021 **Learning the value function from trajectory optimization, using Sobolev descent.**, *LAAS-CNRS, Gepetto*, Toulouse, FRANCE
Under the supervision of Nicolas MANSARD.
○ Keywords : Robotics, Reinforcement Learning, Optimal Control, Differential Dynamic Programming (DDP)
- 11/2019–08/2020 **Routine Bandits : Regret Minimization in Recurrent Problems**, *Université Laval*, Québec, QC, CANADA
Under the supervision of Audrey DURAND.
○ Design of a Transfer Learning Algorithm, Theoretical Validation, Validation through Numerical Experiments.
- 04/2019–09/2019 **Minimization of Supervision for Incremental Deep Learning**, *Commissariat à l'Energie Atomique*, Gif-sur-Yvette, FRANCE
Under the supervision of Adrian POPESCU.
○ Study of Incremental Learning for Image Classification, Design of Semi-Supervised Algorithms.
- 04/2018–07/2018 **Method for Simulating Post-Bayesian Log-Concave Distributions via Langevin Diffusion**, *IMT*, Toulouse, FRANCE
Under the supervision of Sébastien GADAT and Clément PELLEGRINI.
○ Estimation of the Posterior Mean via Langevin Markov Diffusion.
- 01/2017–06/2017 **Towards High-Dimensional Stochastic Learning**, *Centre de Mathématiques et de Leurs Applications*, ENS Cachan, FRANCE
Under the supervision of Alain TROUVE and Luba TCHERTANOV.
○ Modeling the Conformational Dynamics of Molecules with Mathematical Programming (Python).

Teaching Experience

- 09/2016– **Private Mathematics Tutoring: Support, Remedial Classes, Homework Assistance**, *From High School Level to Bachelor's Level*, Paris, FRANCE

Publications

- [1] Amit Parag, Sébastien Kleff, Léo Saci, Nicolas Mansard, and Olivier Stasse. Value learning from trajectory optimization and sobolev descent: A step toward reinforcement learning with superlinear convergence properties. 2022.
- [2] Hassan Saber, Léo Saci, Odalric-Ambrym Maillard, and Audrey Durand. Routine bandits: Minimizing regret on recurring problems. 2021.

Interests

Probability, Statistics, Stochastic Calculus, Programming, Machine Learning, Mathematical Modeling

Language Skills

French Native Language
English Certified B2 Level, IELTS Academic (2018)

Computer Skills

Python Scikit-learn, Pytorch, Keras, Pandas, Gym
R Advanced Skills
VBA, SQL Basic Skills
Git Basic Skills